CNN Architecture

Outline

Classic Architecture
- AlexNet
- VGG
- GoogLeNet
- ResNet

More Examples
- Improving ResNet
- Various Architecture

Summary

References

- Deep Learning by Goodfellow, Bengio and Courville
  - Chapter 9

- Online resources:
  - Deep Learning Specialization (coursera)
  - Stanford CS231n: CNN for Visual Recognition
  - Dive into Deep Learning

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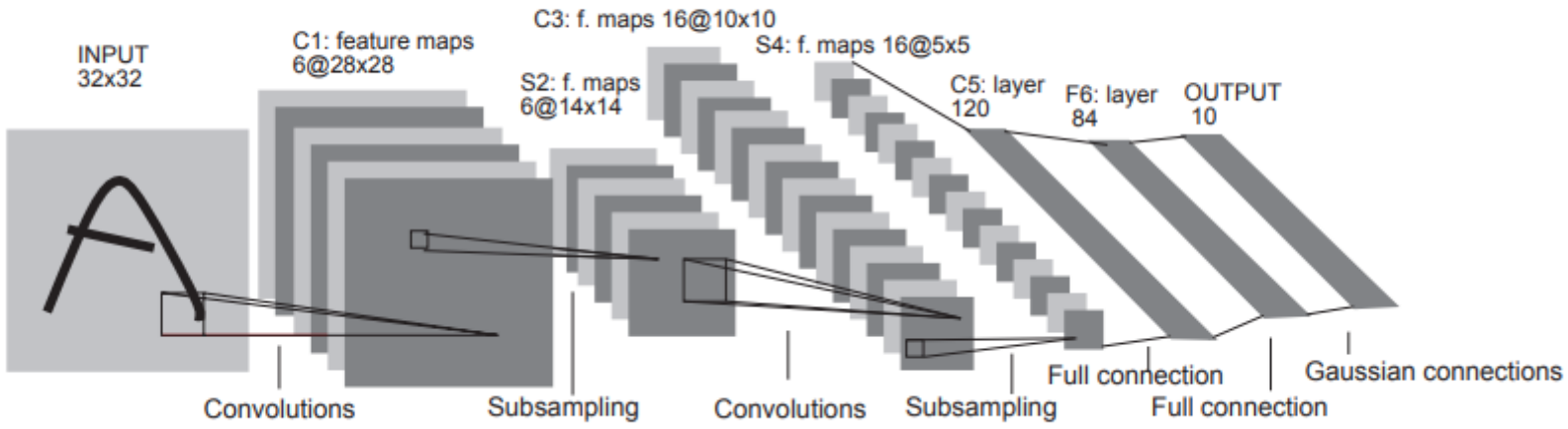
Classic: LeNet-5



- CONV-POOL-CONV-POOL-FC-FC
  - 5 x 5 conv filters (stride 1)
  - 2 x 2 pooling layers (stride 2)

Image source: Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," Proceedings of the IEEE, vol. 86, no. 11, pp. 2278–2324, 1998

ImageNet Challenge winners

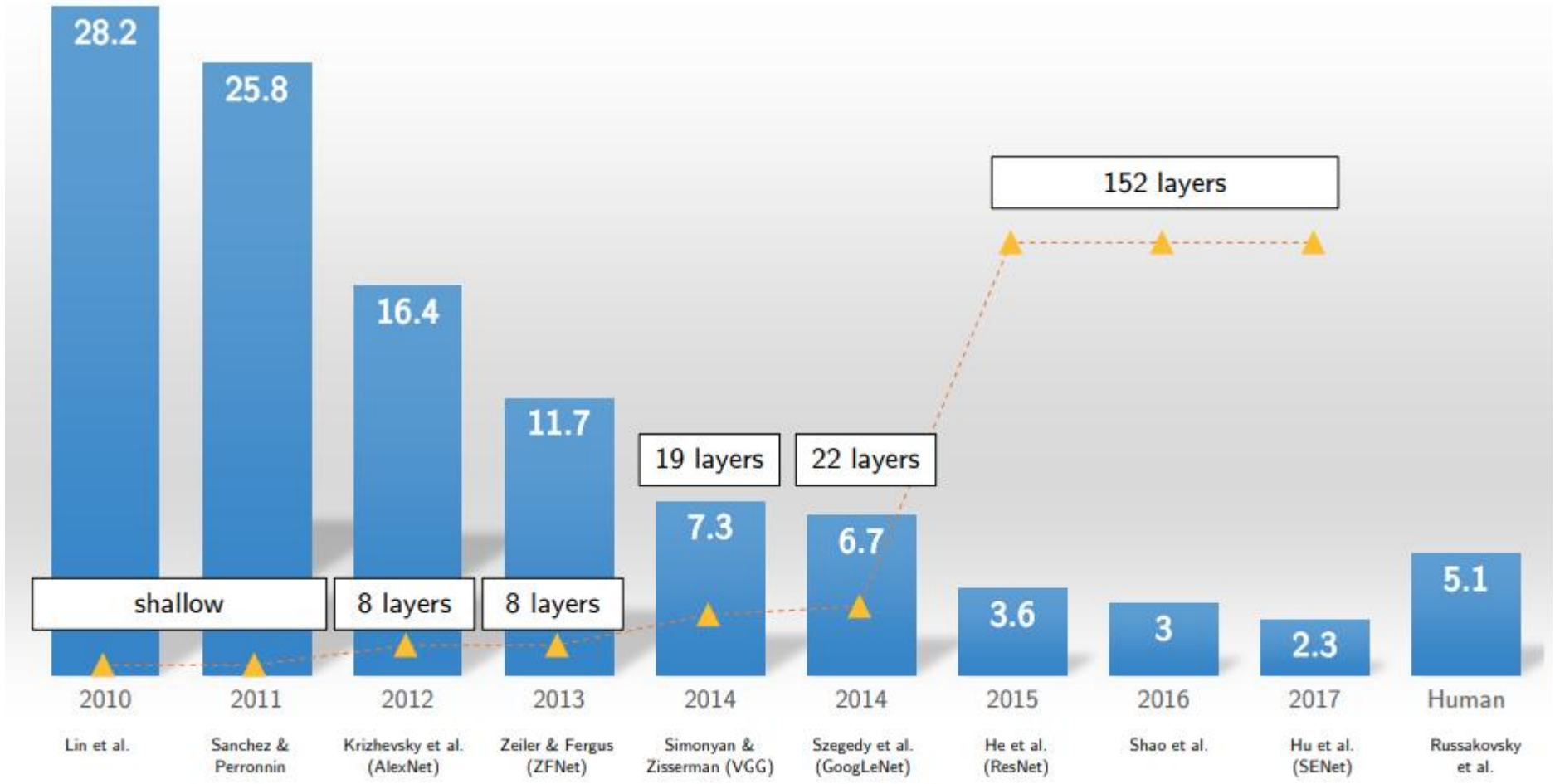


Image modified from: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.htm

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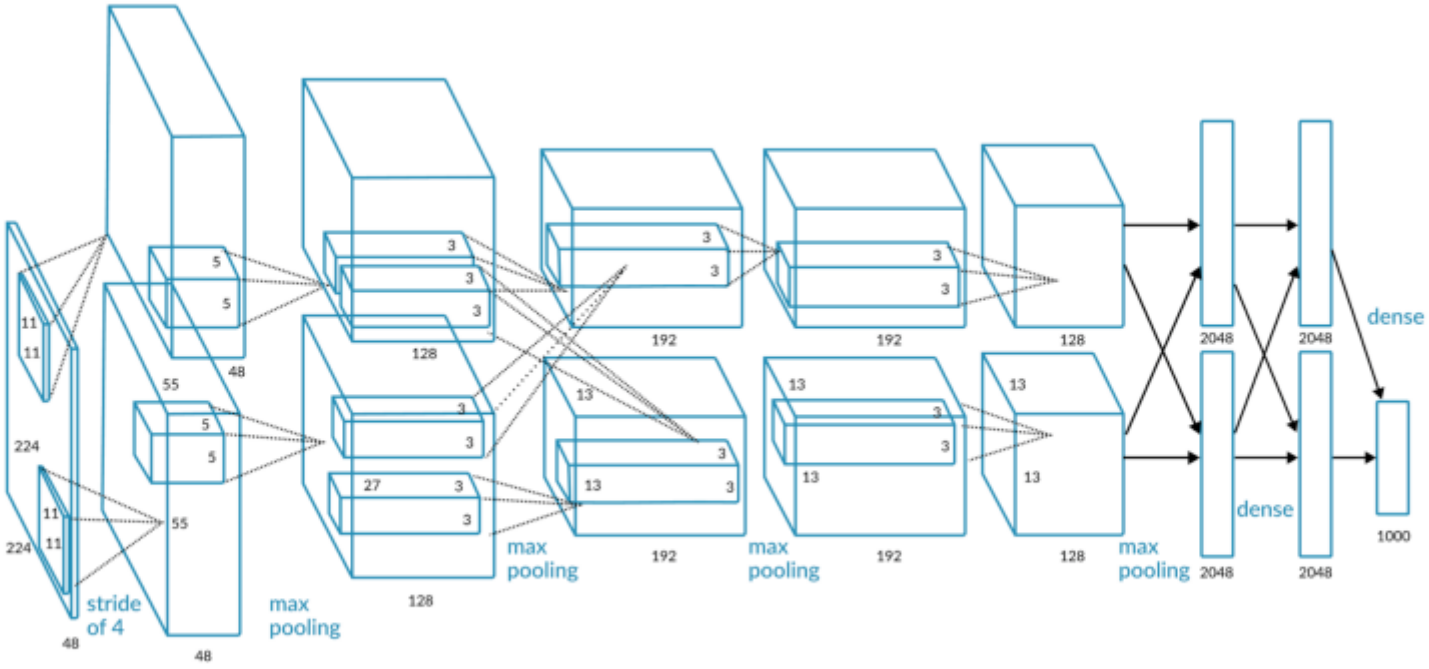
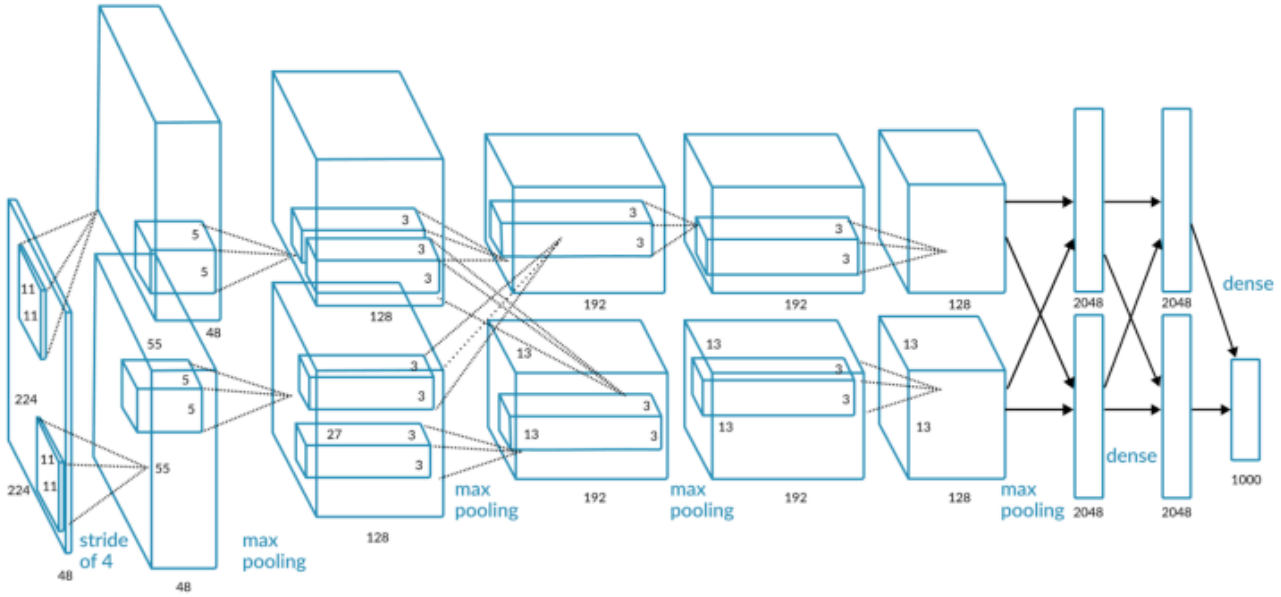
AlexNet



- Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton (2012)
  - ILSVRC 2012 winnter

image source: https://missinglink.ai/guides/convolutional-neural-networks/ convolutional-neural-network-architecture-forging-pathways-future/ (modified from A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in Advances in Neural Information Processing Systems, pp. 1097–1105, 2012)

AlexNet



227x227x3 INPUT

55x55x96 CONV1 96 11x11 filters at stride 4, pad 0
27x27x96 MAX POOL1 3x3 filters at stride 2
27x27x96 NORM1 normalization layer

27x27x256 CONV2 256 5x5 filters at stride 1, pad 2
13x13x256 MAX POOL2 3x3 filters at stride 2
13x13x256 NORM2 normalization layer

13x13x384 CONV3 384 3x3 filters at stride 1, pad 1
13x13x384 CONV4 384 3x3 filters at stride 1, pad 1
13x13x256 CONV5 256 3x3 filters at stride 1, pad 1
6x6x256 MAX POOL3 3x3 filters at stride 2

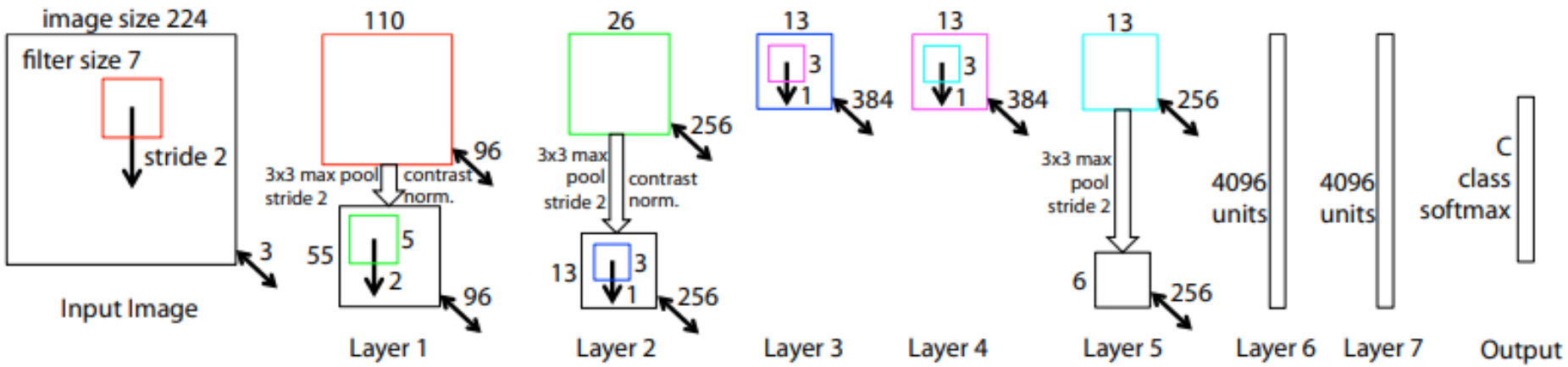
4096 FC6 4096 neurons
4096 FC7 4096 neurons
1000 FC8 1000 neurons (class scores)

Image source: A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in Advances in Neural Information Processing Systems, pp. 1097–1105, 2012

- Details:
  - first use of ReLU
  - used normalization (NORM) layers (not common anymore)
  - heavy data augmentation
  - dropout: 0.5
  - batch size: 128
  - SGD + momentum (0.9)
  - learning rate: 10^{-2} (reduce by 10 manually when validation accuracy plateaus)
  - L_2 weight decay: 5×10^{-4}
  - 7 CNN ensemble: 18.2% -> 15.4%

- traind on GTX 580 GPU (3GB memory)
  - network spread across 2 GPUs

ZFNet (Zeiler and Fergus, 2013)



- the same as AlexNet but
  - CONV1: change from(11x11 stride 4) to (7x7 stride 2)
  - CONV 3, 4, 5: instead of 384, 384, 256 filters use 512, 1024, 512
  - ImageNet top 5 error: 16.4% -> 11.7%

image source:Zeiler, Matthew D., and Rob Fergus. "Visualizing and understanding convolutional networks." European Conference on Computer Vision. Springer, Cham, 2014

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VGG

- Simonyan and Zisserman (2014)
- key idea: small filters, deeper networks
  - Only

    3x3 CONV stride 1, pad 1

    2x2 MAX POOL stride 2
- ILSVRC top 5 error
  - 11.7% (ZFNet, 2013) -> 7.3% (VGG, 2014)
- two versions: VGG16, VGG19
  - VGG19 only slightly better

    (use more memory)

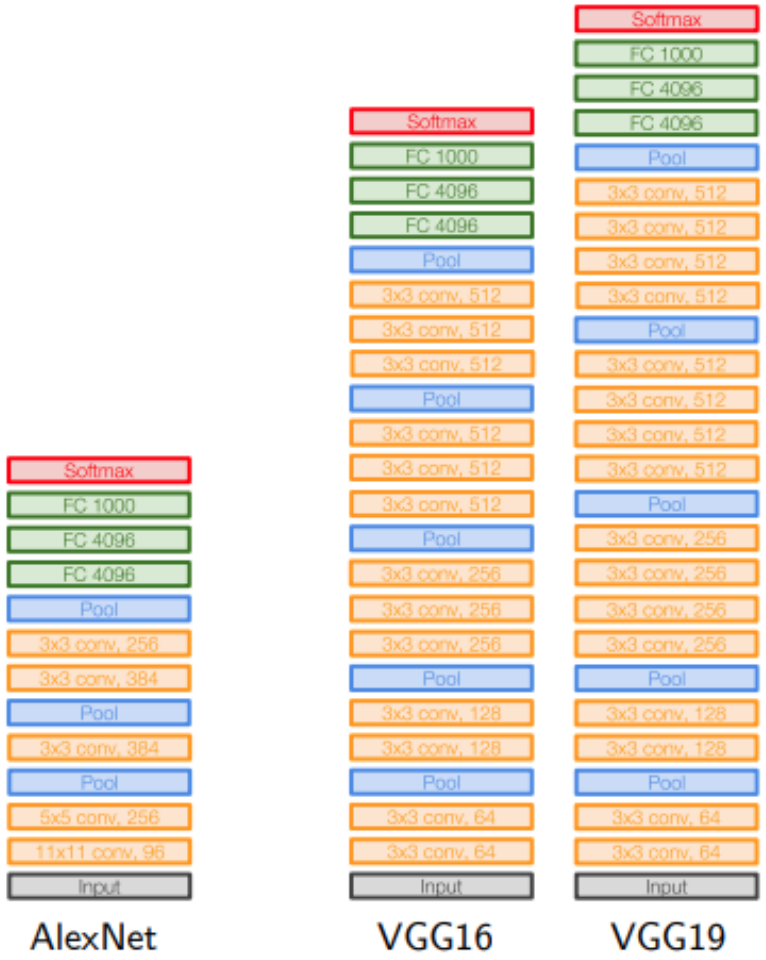


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Why use smaller filter?

- consider stacking three 3x3 conv(stride 1) layers

- benefits
  - its effective receptive field
    - that of one 7x7 conv layer
  - but deeper
    - more non-linearities
  - fewer parameters

    $3 \times (3^2C^2)$ vs 7^2C^2

    where C is the number of channels(filters) per layer

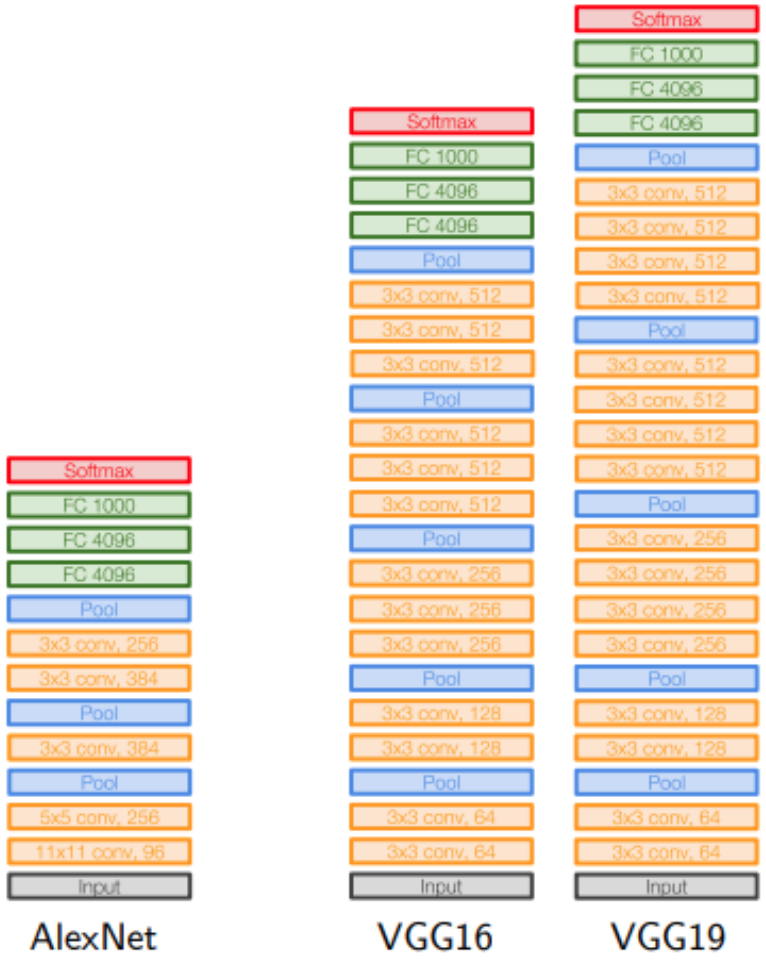
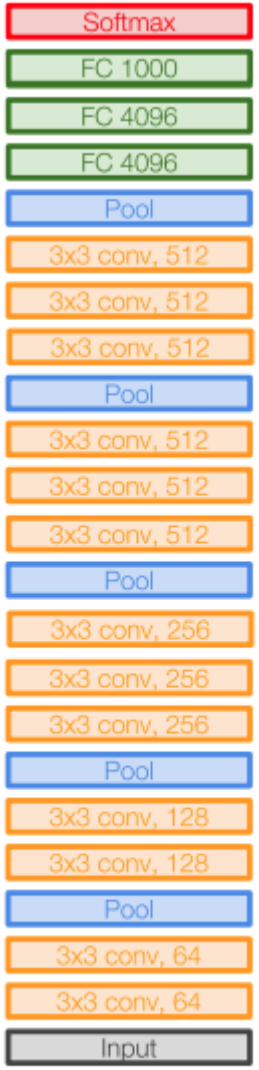


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Architecture (VGG 16)

layer type	dimension	memory (96MB/image)	parameters (138M total)
INPUT	224x224x3	224*224*3=150K	0
CONV3-64	224x224x64	224*224*64=3.2M	(3*3*3)*64 = 1,728
CONV3-64	224x224x64	224*224*64=3.2M	(3*3*64)*64 = 36,864
POOL2	112x112x64	112*112*64=800K	0
CONV3-128	112x112x128	112*112*128=1.6M	(3*3*64)*128 = 73,728
CONV3-128	112x112x128	112*112*128=1.6M	(3*3*128)*128 = 147,456
POOL2	56x56x128	56*56*128=400K	0
CONV3-256	56x56x256	56*56*256=800K	(3*3*128)*256 = 294,912
CONV3-256	56x56x256	56*56*256=800K	(3*3*256)*256 = 589,824
CONV3-256	56x56x256	56*56*256=800K	(3*3*256)*256 = 589,824
POOL2	28x28x256	28*28*256=200K	0
CONV3-512	28x28x512	28*28*512=400K	(3*3*256)*512 = 1,179,648
CONV3-512	28x28x512	28*28*512=400K	(3*3*512)*512 = 2,359,296
CONV3-512	28x28x512	28*28*512=400K	(3*3*512)*512 = 2,359,296
POOL2	14x14x512	14*14*512=100K	0
CONV3-512	14x14x512	14*14*512=100K	(3*3*512)*512 = 2,359,296
CONV3-512	14x14x512	14*14*512=100K	(3*3*512)*512 = 2,359,296
CONV3-512	14x14x512	14*14*512=100K	(3*3*512)*512 = 2,359,296
POOL2	7x7x512	7*7*512=25K	0
FC	1x1x4096	4096	7*7*512*4096 = 102,760,448
FC	1x1x4096	4096	4096*4096 = 16,777,216
FC	1x1x1000	1000	4096*1000 = 4,096,000



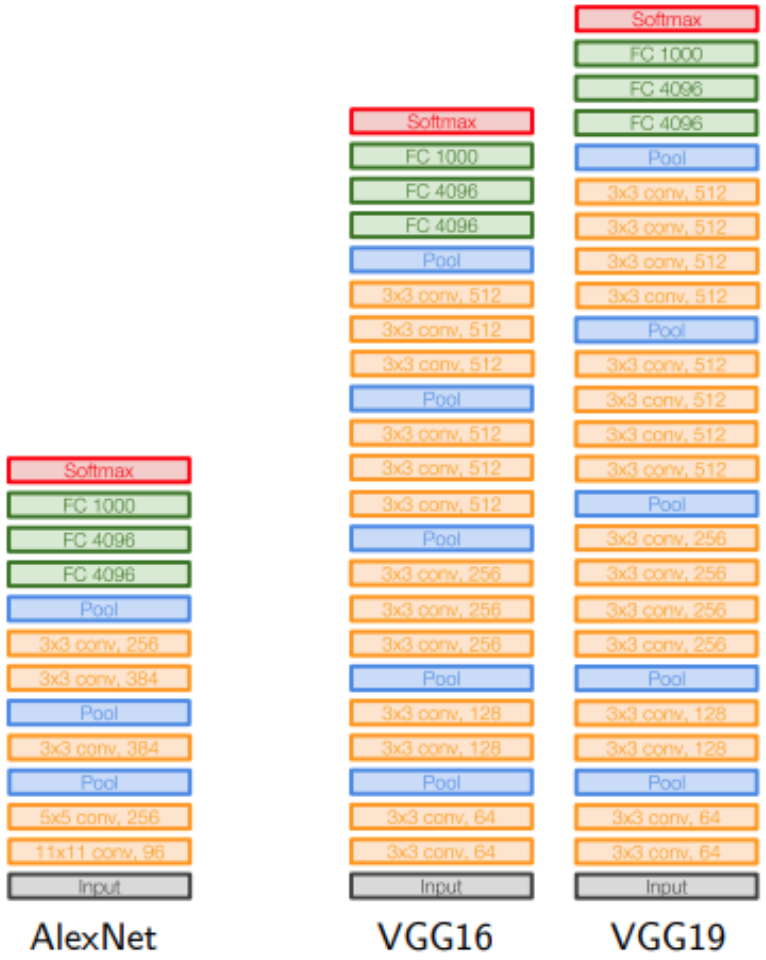
VGG16

- most memory: in early CONV
- most parameters: in late FC

image and table source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.htm

- ILSVRC'14 ranking: 2nd in classification, 1st in localization

- details:
  - similar training procedure as AlexNet
  - no local response normalization (LRN)
  - FC7 features generalize well to other tasks



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GoogLeNet

- Szegedt et al. (2014)

- key idea: deeper networks with computational efficiency
  - 22 layers
  - efficient “inception” module
  - minimal use of FC layers
  - only 5 million parameters

    (12x less than AlexNet)
  - ILSVRC’14 classification winner (6.7% top 5 error)

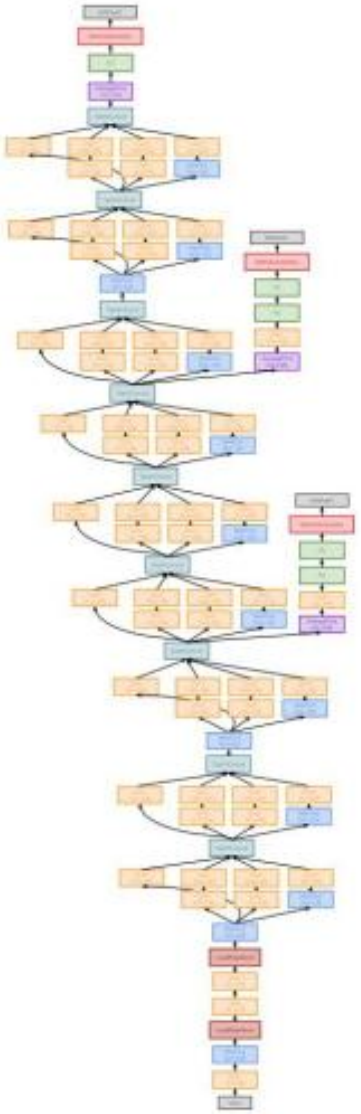
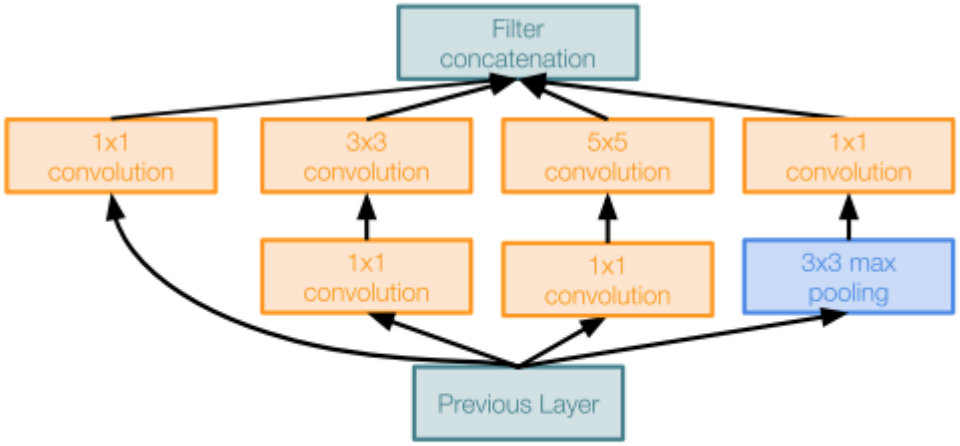


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

GoogLeNet

- inception module
  - design a good local network topology (network within network)
  - stack these modules



inception module

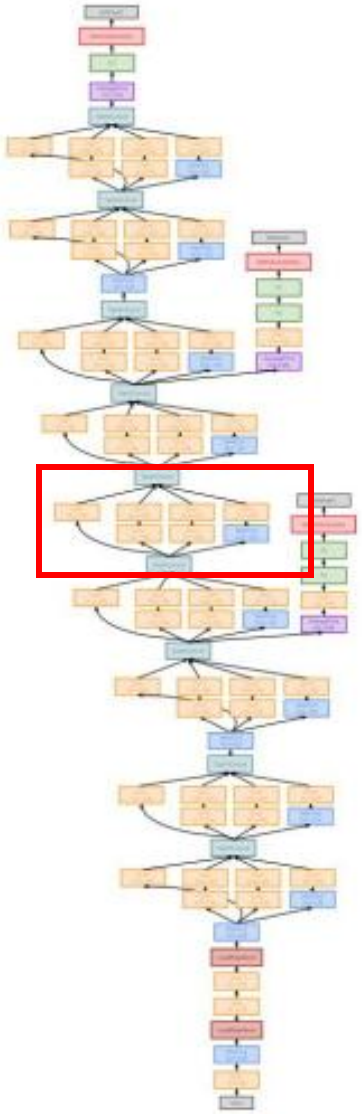
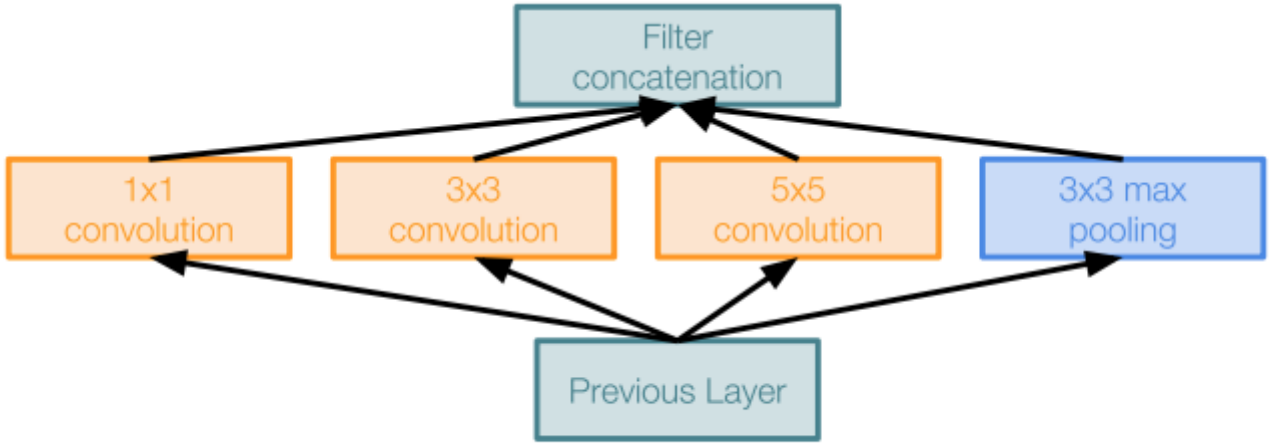


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Naïve inception module

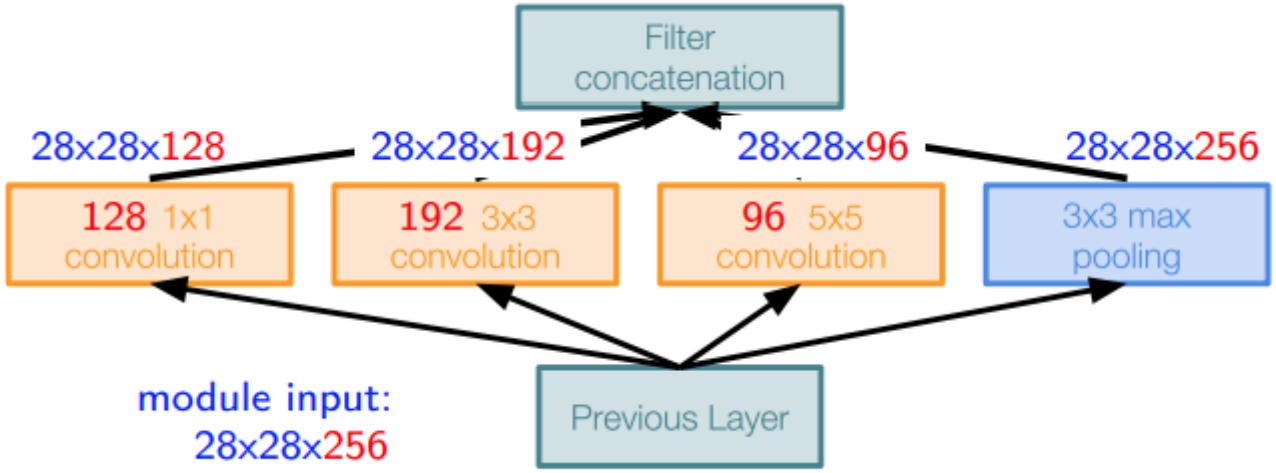


- apply parallel filter operations on the input
  - multiple receptive field sizes (1x1, 3x3, 5x5) for convolution
  - pooling (3x3)
- concatenate all filter outputs together : depth-wise

image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Naïve inception module

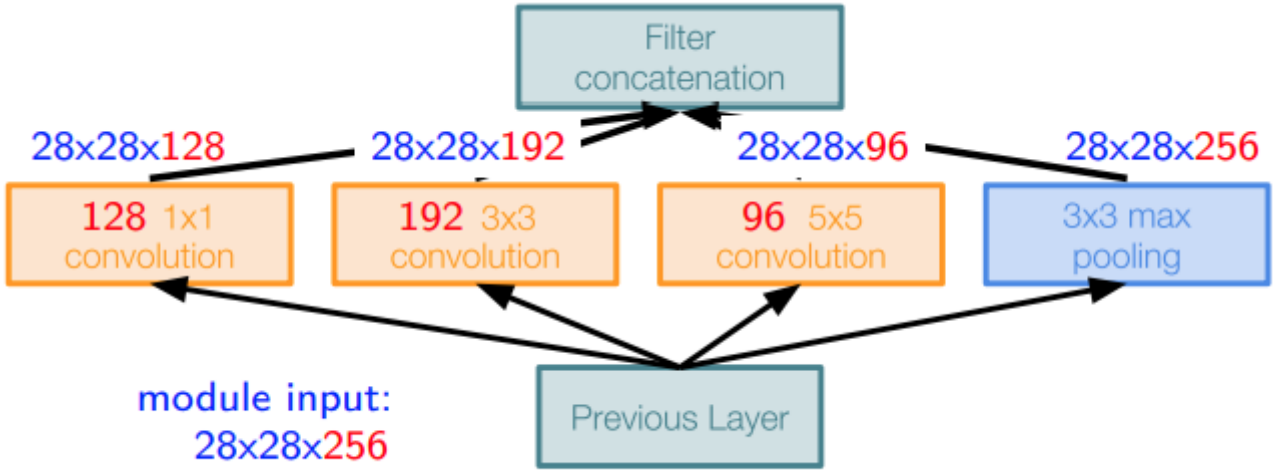
- problem with this idea: compute-intensive



- output size after filter concatenation: 529k
  - 28x28x(128+192+96+256) = 28x28x672
- total number of convolution operations: 854M
  - 28x28x128x1x1x256 + 28x28x192x3x3x256 + 28x28x96x5x5x256
  - very expensive to compute

image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

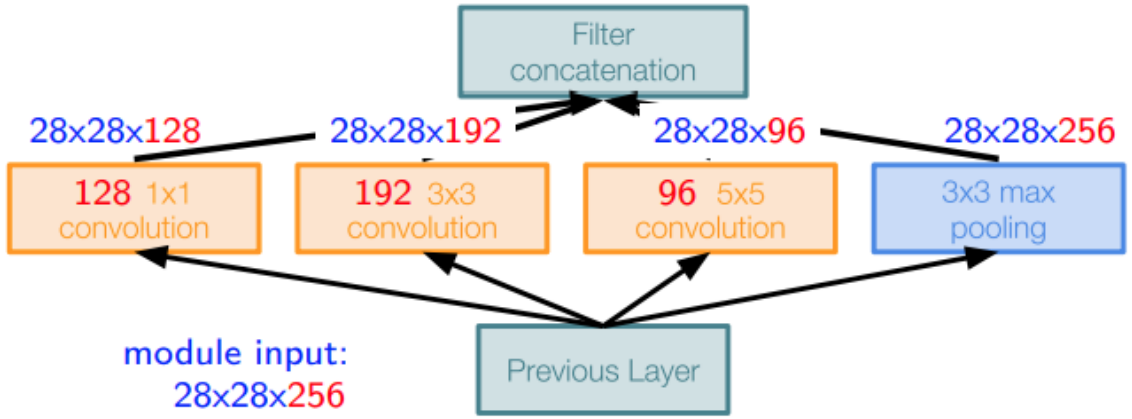
Naïve inception module



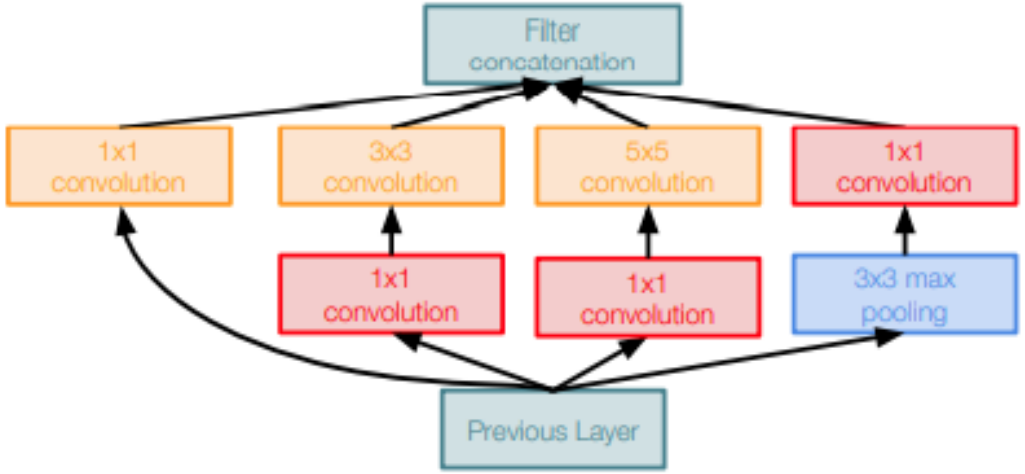
- another challenge:
  - pooling layer preserves feature depth
  - total depth after concatenation can only grow at every layer
- solution:
  - bottleneck layers (use 1x1 convolution to reduce feature depth)

image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Naïve inception module



inception module with dim reduction



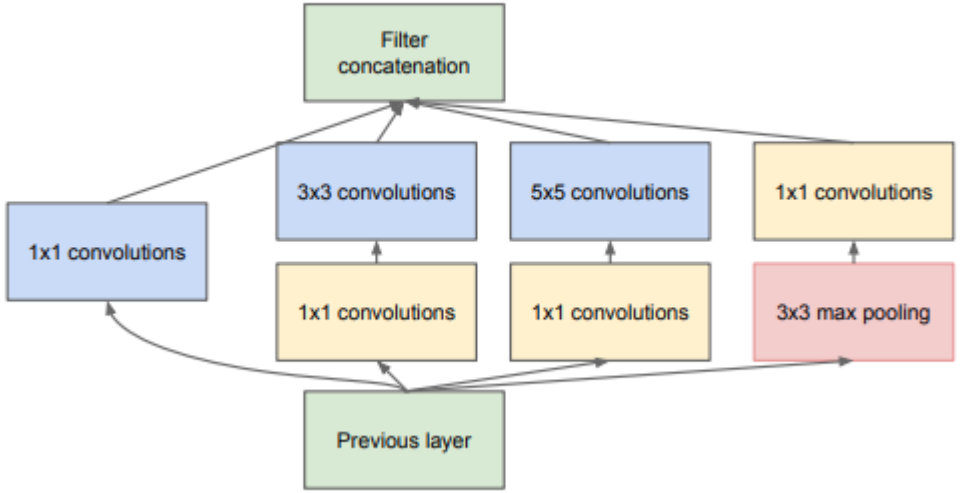
- 1x1 conv "bottleneck" layers
- the same setup as on page 20

  854M operations -> 358M operations

image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

GoogLeNet

- stacked inception modules
  - with dimension reduction on top of each other



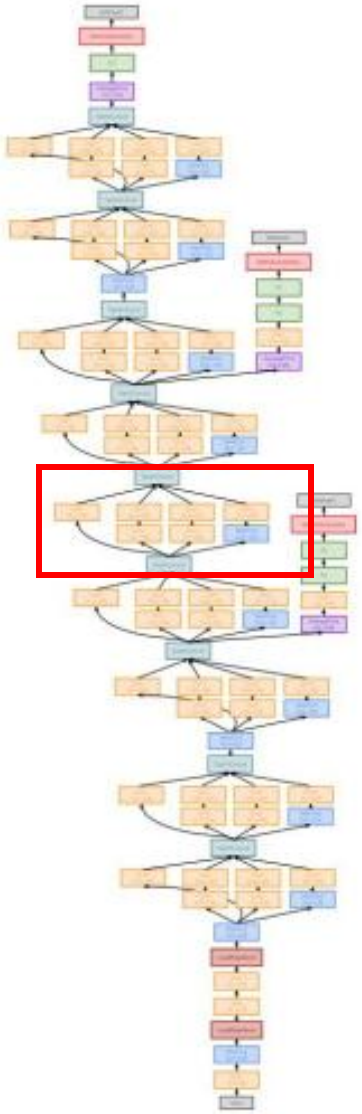
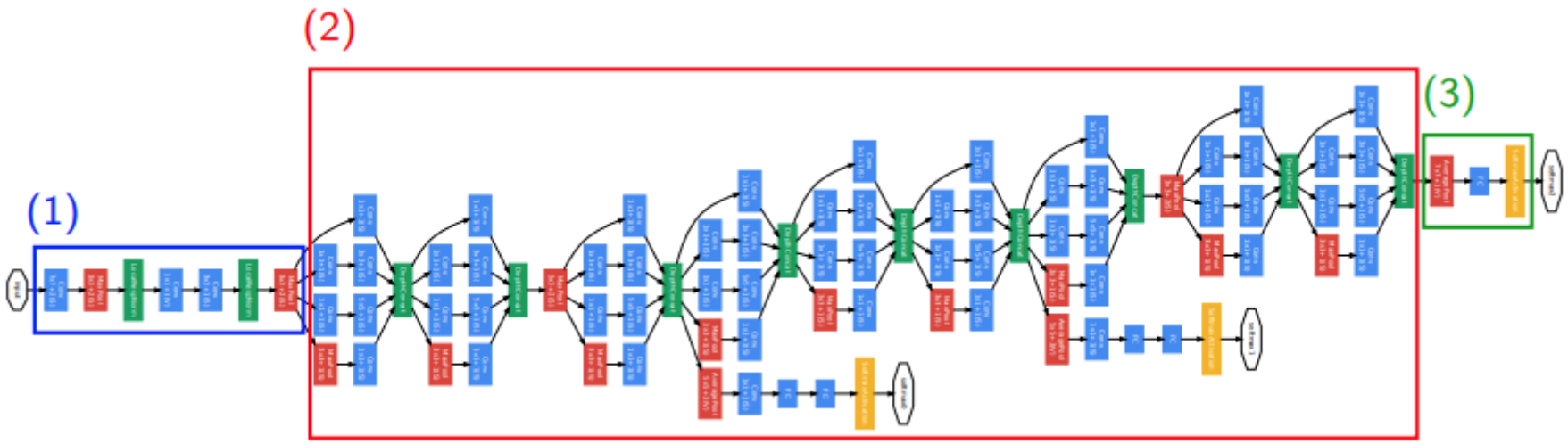


image source: C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015

GoogLeNet

- full architecture

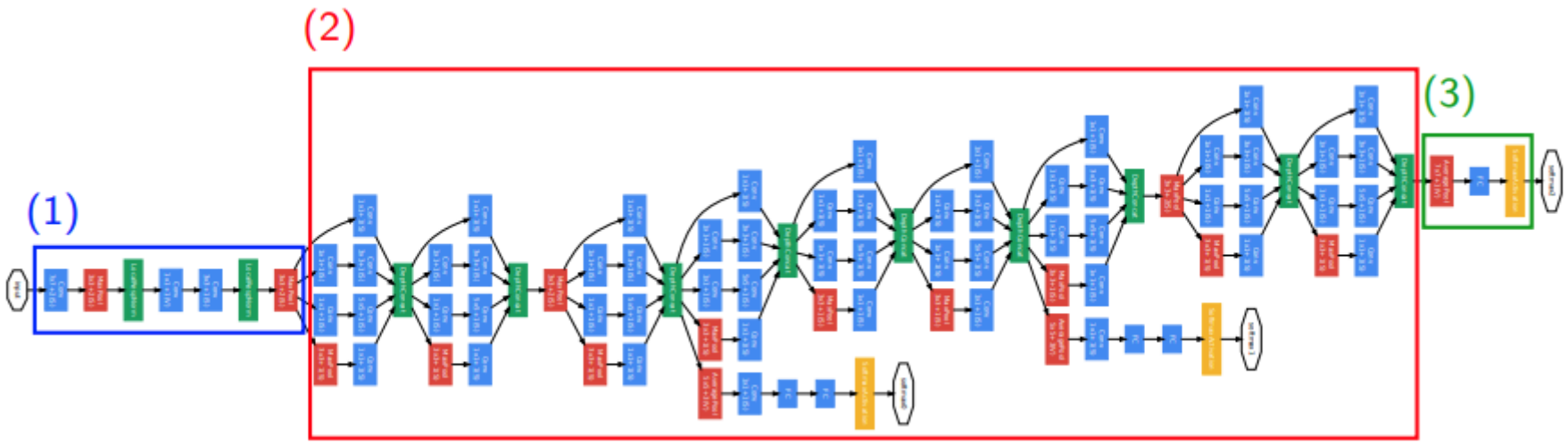


- (1) stem network: CONV-POOL-2xCONV-POOL
- (2) stacked inception modules
- (3) classifier output

image source: C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015

GoogLeNet

- full architecture



  - (4) auxiliary classification outputs: AvgPOOL-1x1CONV-FC-FC-SOFTMAX
- total 22 layers with weights
  - parallel layers count as 1 layer: 2 layers per inception module
  - auxiliary output layers: not counted

image source: C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015

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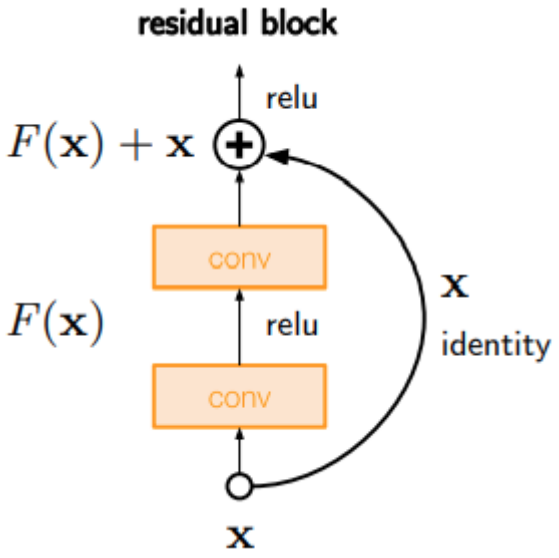
ResNet

- He et al. (2015)

- key idea: very deep nets using residual connections
  - 152 layers model for ImageNet
  - ILSVRC'15 classification winner (3.57% top 5 error)



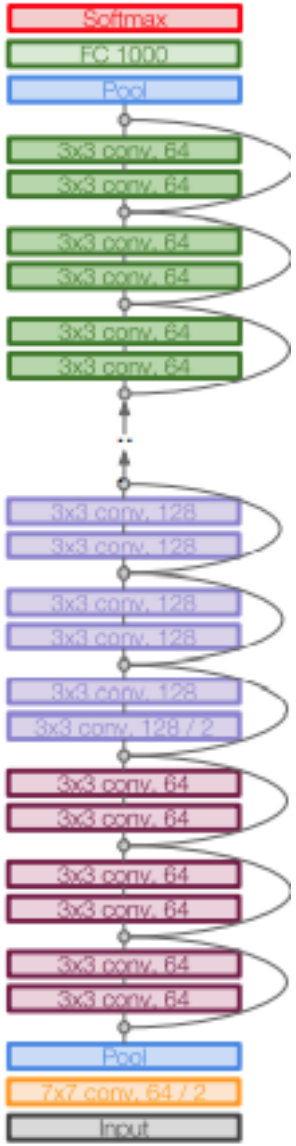


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

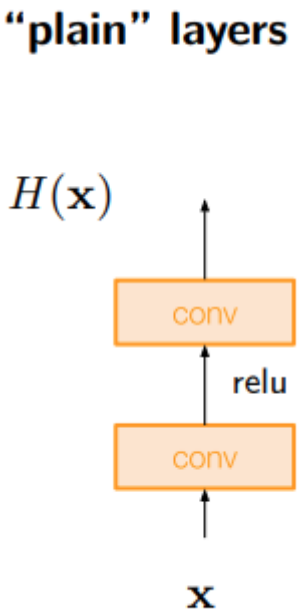
ResNet

- intuition
  - if trained appropriately, deeper models should be able to perform at least as well as shallower models

- a solution by construction:
  - copy the learned layers from the shallower model
  - set additional layers to identity mapping

Residual Block

- use network layers to fit a residual mapping
  - residual mapping: $F(x) = H(x) - x$
  - instead of directly trying to fit a desired underlying mapping $H(x)$



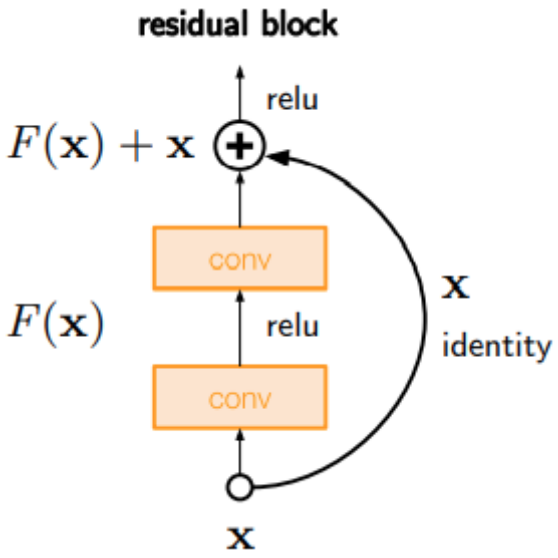


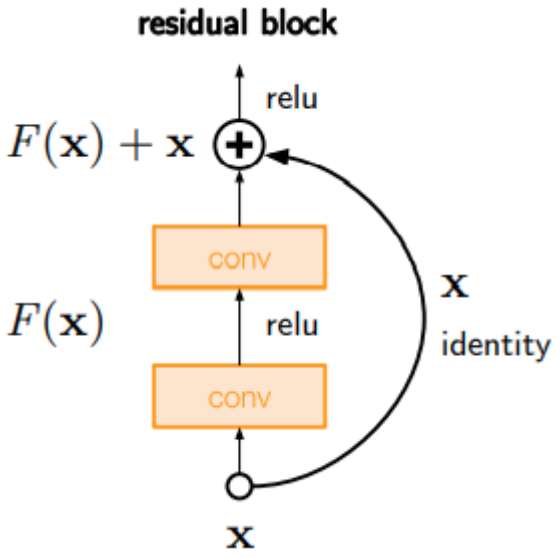
image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

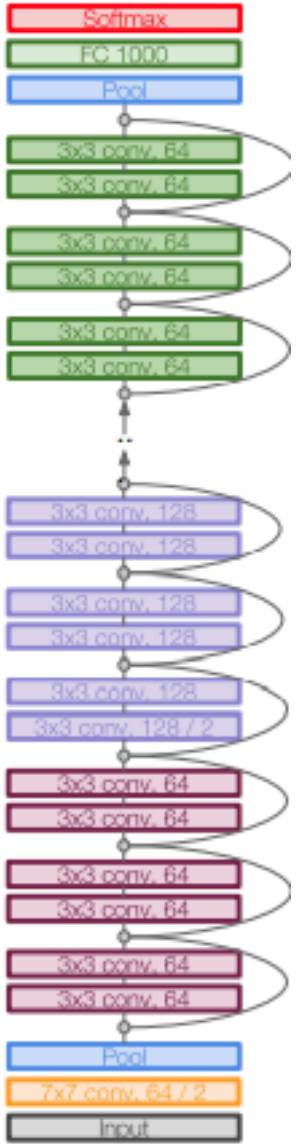
Residual Block

- 1. adds interpretation to L2 regularization in conv net context
  - set all weights in RB to zero -> it computes identity
    - it is easy for the model to learn not to use the layers that it does not need
  - driving parameters toward zero
    - in standard conv nets: makes no sense
    - in ResNet: encourages the model not to learn the layers not needed
- 2. has nice gradient flow in backward pass
  - residual connections = gradient super highway
    - much easier/faster training
  - managing gradient flow
    - very important in ML: will be revisited many times

ResNet Architecture

- stack residual blocks
- every residual block
  - has two 3x3 conv layers
- periodically
  - double the number of filters
  - downsample spatially (stride 2)
- at the beginning
  - additional conv layer
- no FC layers at the end
  - only FC1000 to output classes





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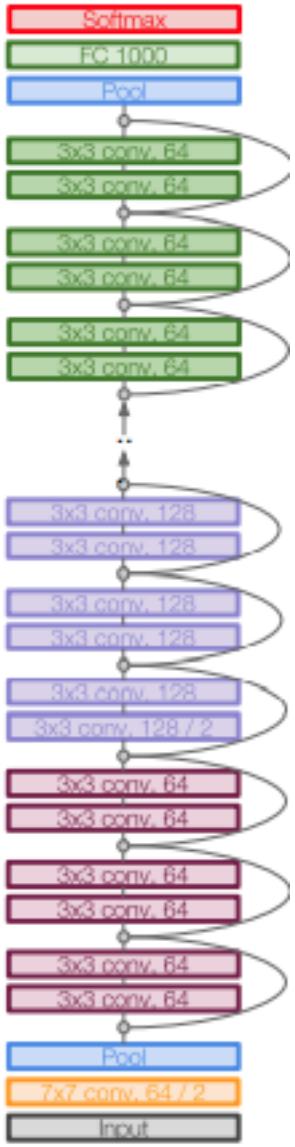
ResNet Architecture

- total depths:
  - 34, 50, 101 or 152 layers for ImageNet



- for deeper nets (50 + layers)
  - use bottleneck layers to improve efficiency

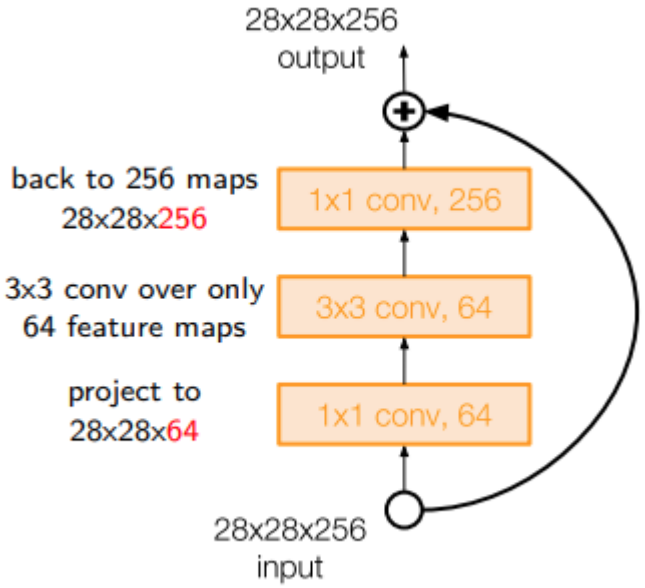
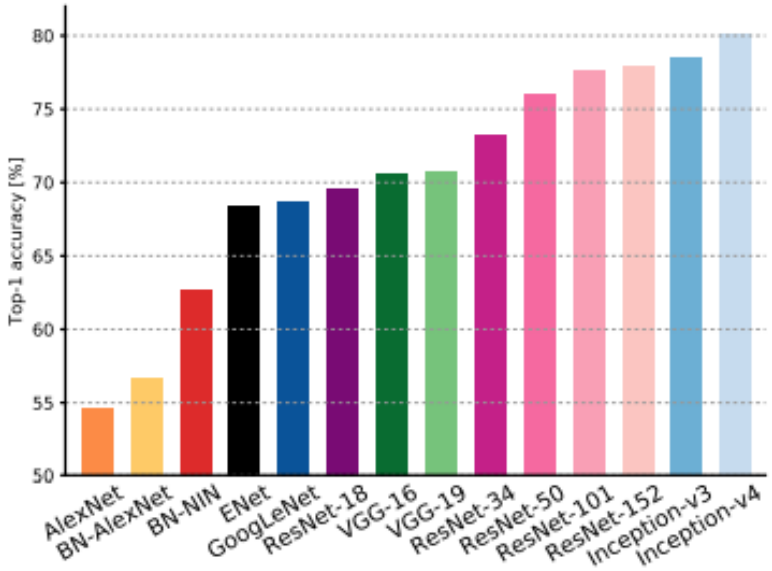
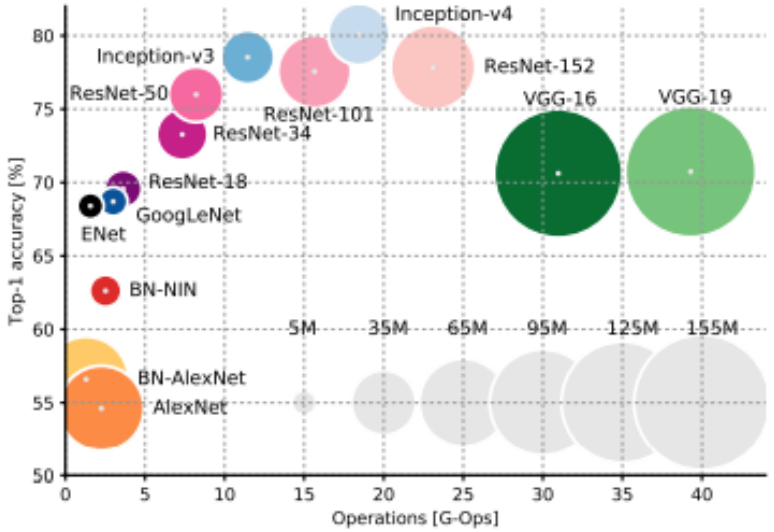


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

- training details
  - batch normalization after every CONV layer
  - Xavier initialization: initial weight $\sim \mathcal{N}\left(0, \frac{1}{n}\right)$ where n is the number of neurons
  - SGD + momentum (0.9)
  - learning rate: 0.1 (divided by 10 when validation error plateaus)
  - mini batch size: 256
  - weight decay: 10^{-5}
- results
  - ILSVRC'15 winner in all five main tracks (3.6% top 5 error)
  - better than "human" performance (Russakovsky, 2014)

Comparison





- Inception
- VGG: highest memory, most operations
- GoogLeNet: most efficient
- AlexNet: smaller compute, still memory heavy, lower accuracy
- ResNet: moderate efficiency depending on model, highest accuracy

x-axis: the amount of operations for a single forward pass; circle size ∝ the number of parameters

Image Source:Canziani, Alfredo, Adam Paszke, and Eugenio Culurciello. "An analysis of deep neural network mdodels for practical applications" arXiv prerping arXiv:1605.07678 (2016)

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Classic Architecture
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Summary

Improving ResNets

- ideas:
  - improved residual block
  - wide ResNet
  - ResNeXt
  - stochastic depth
  - multi-scale ensembling
  - feature recalibration (SENet)

Identity mappings in deep residual networks (He et al., 2016)

- improved ResNet block design
  - creates a more direct path for propagating info throughout net
  - moves activation to residual mapping pathway

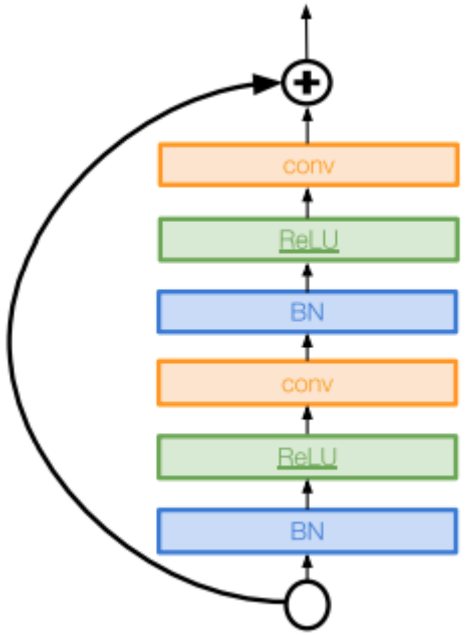


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Wide ResNet (Zagoruyko et al., 2016)

- authors argue: residuals are the important factor, not depth
- use wider residual blocks
  - $F \times K$ filters instead of F filters in each layer
  - 50-layer wide ResNet outperforms 152-layers (original ResNet)
- computational benefit
  - increasing width (instead of depth)
  - more computationally efficient (parallelizable)

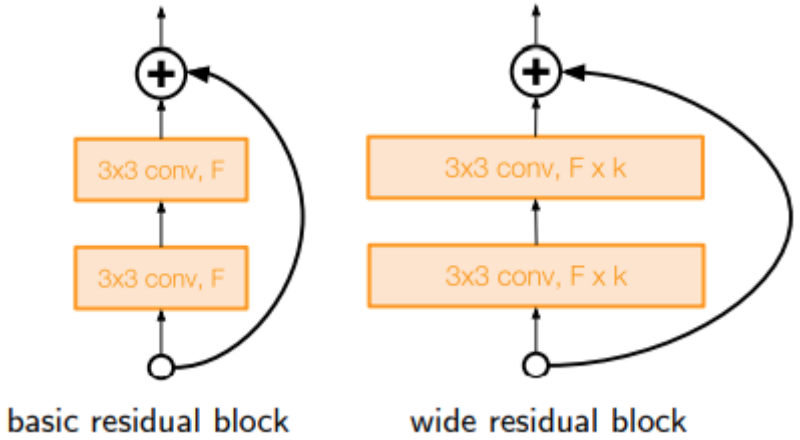


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

ResNeXt (Xie et al., 2016)

- aggregated residual transformations
  - increases width of residual block through multiple parallel pathways
  - similar in spirit to inception module

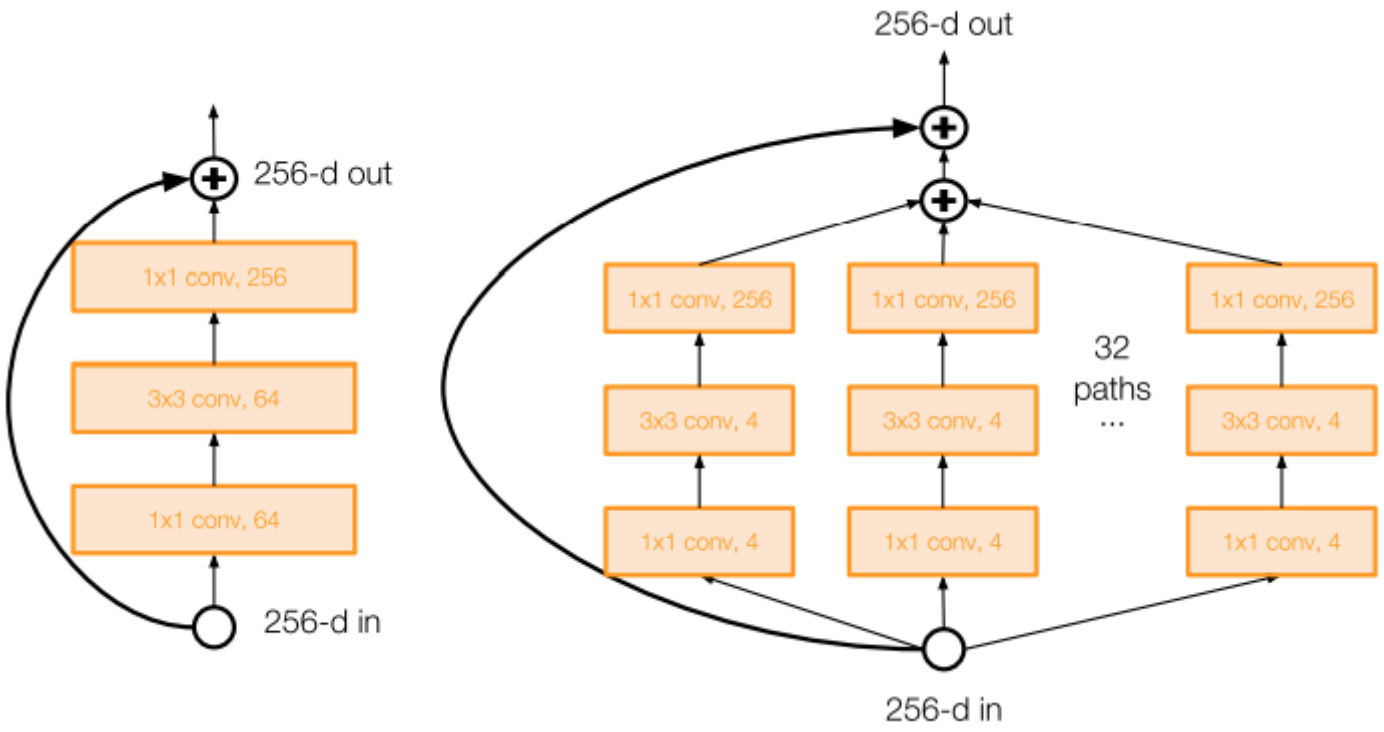


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Stochastic depth (Huang et al., 2016)

- motivation:
  - reduce vanishing gradients and training time through short networks during training
- details:
  - randomly drop a subset of layers during each training pass
  - bypass with identity function
  - use full deep network at test time

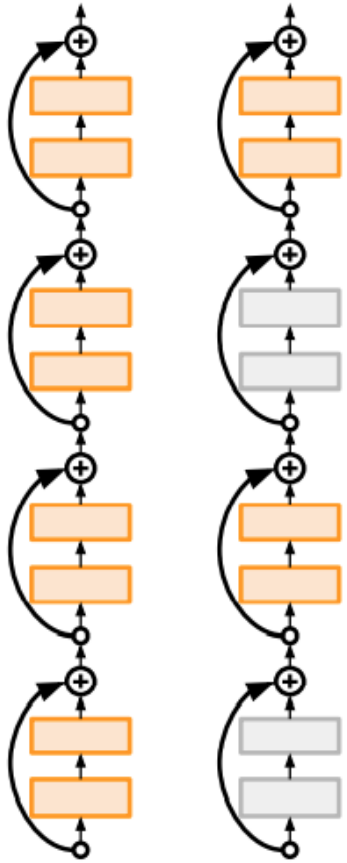
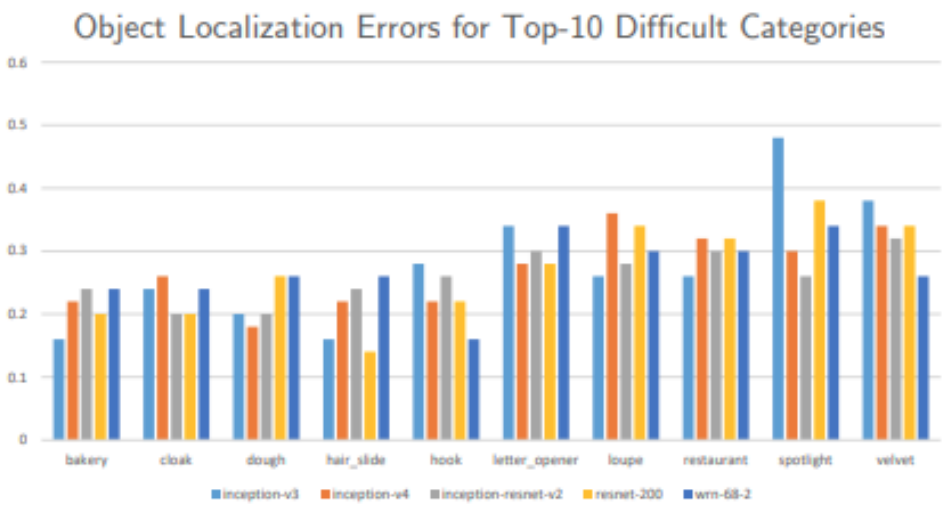


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

Multi-scale ensembling (Shao et al., 2016)

- ILSVRC'16 classification winner
  - "Good Practices for Deep Feature Fusion"
- idea: multi-scale ensembling of
  - inception, inception-ResNet, ResNet, wide ResNet models



method	error (%)
Resnet-200	4.26
Inception-v3	4.20
Inception-v4	4.01
Inception-Resnet-v2	3.52
Fusion (val)	2.92
Fusion (test)	2.99

image source: Shao, Jie, et al. "Good pratices for deep feature fusion." European Conference on Computer Vision, 2016. http://image-net.org/challenges/talks/2016/Trimps Soushen@ILSVRC2016.pdf

Squeeze-and-Excitation Networks (SENet) (Hu et al., 2018)

- ILSVRC'17 classification winner
  - base architecture: ResNeXt-152
  - introduces SE block (applicable to a variety of nets)
  - squeeze: global information embedding
  - excitation: adaptive recalibration
- main idea:
  - improve representational power of a network by modeling interdependencies between channels of conv features

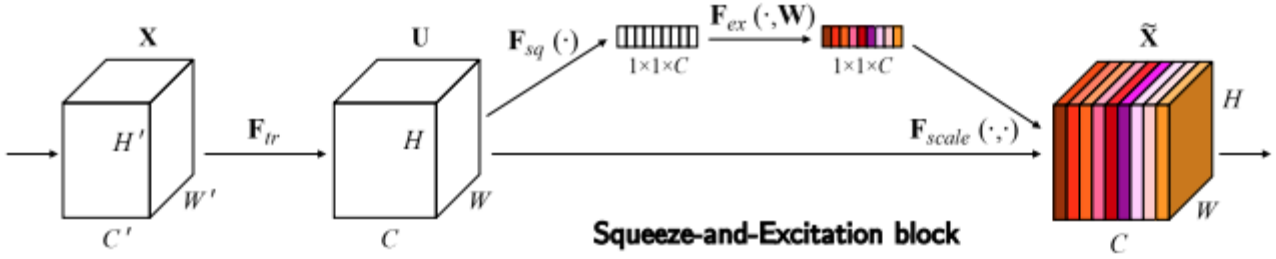
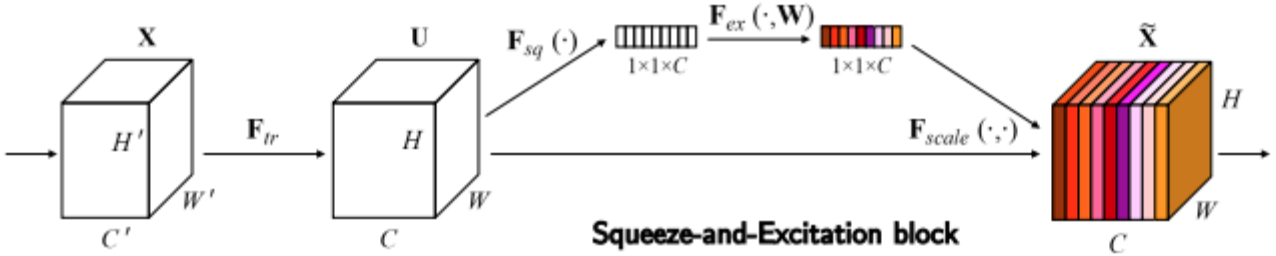


image source: Hu, Jie, Li Shen, and Gang Sun. "Squeeze-and-excitation networks." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2018



- $\mathbf{F}_{tr} : \mathbf{X}^{H' \times W' \times C'} \mapsto \mathbf{U}^{H \times W \times C}$ (a conv operation)
- $\mathbf{F}_{sq} : \mathbf{U}^{H \times W \times C} \mapsto \mathbf{Z}^{1 \times 1 \times C}$
  - global average pooling (each feature map → a scalar; depth maintained)
  - "global information embedding" (squeeze)
- $\mathbf{F}_{ex} : \mathbf{Z}^{1 \times 1 \times C} \mapsto \mathbf{S}^{1 \times 1 \times C}$
  - $\underbrace{\text{FC} \rightarrow \text{ReLU}}_{\text{compress}} \rightarrow \underbrace{\text{FC} \rightarrow \text{Sigmoid}}_{\text{decompress}}$
  - to calculate scale for each feature map
- $\mathbf{F}_{scale} = \mathbf{S}^{1 \times 1 \times C} \odot \mathbf{U}^{H \times W \times C} = \widetilde{\mathbf{X}}$
  - reweight feature maps
  - "adaptive feature recalibration" (excitation)

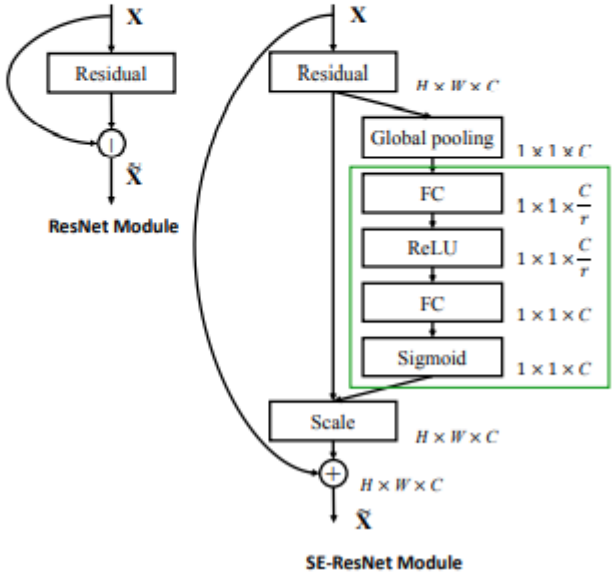


image source: Hu, Jie, Li Shen, and Gang Sun. "Squeeze-and-excitation networks." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2018

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Summary

Directions of developments

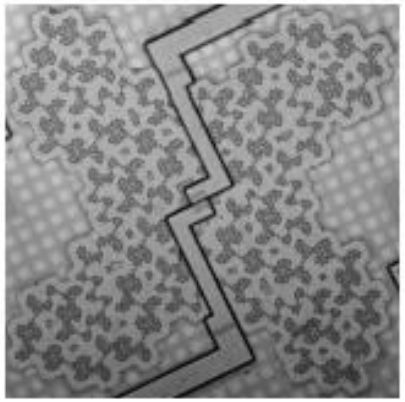
- beyond ResNets:
  - ultra-deep neural networks without residuals (FractalNet)
  - densely connected CNNs (DenseNet)
- efficient networks:
  - Caffe2Go (Facebook), TensorRT (NVIDIA), Core ML (Apple)
  - SqueezeNet, MobileNet
- meta/automated learning:
  - neural architecture search (NAS) and efficient NAS (ENAS)
  - Cloud AutoML (Google)

FactalNet (Larsson et al., 2017)

- ultra-deep neural networks without residuals

- argue:
  - key is trasitioning effectively from shallow to deep
  - residual representations are not necessary

- propose: fractal architecture
  - both shallow and deep paths to output
  - trained with dropping out subpaths
  - full network at test time



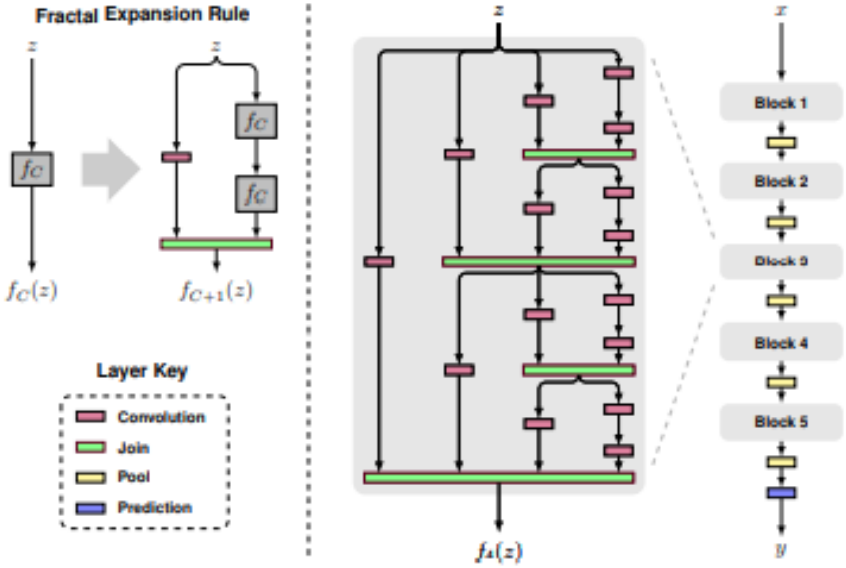


image sources: Samavati, et al. "Fractal capacitors." IEEE Journal of Solid-State Circuits 33.12 (1998): 2035-2041; Larsson et al. "FractalNet: Ultra-deep neural networks without residuals." arXiv preprint arXiv:1605.07648 (2016)

DenseNet (Huang et al., 2017)

- densely connected convolutional networks

- idea: dense blocks
  - each layer is connected to every other layer in feedforward fashion

- benefits:
  - alleviates vanishing gradient
  - strengthens feature propagation
  - encourages feature reuse

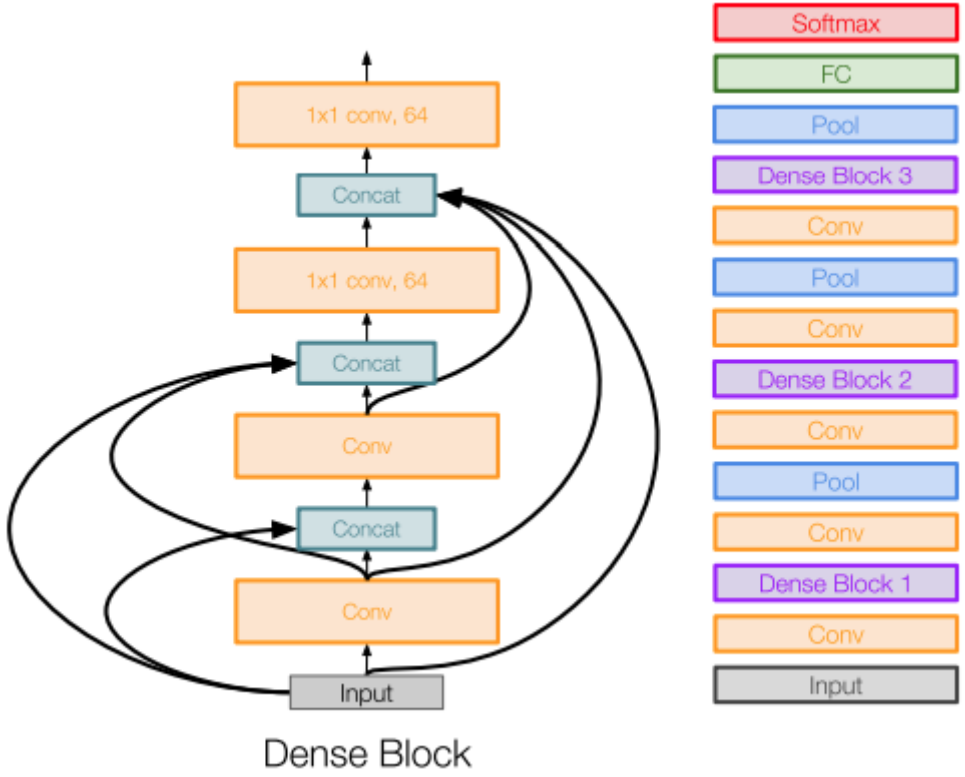


image source: Fei-Fei Li, J. Johnson, S. Yeung, CS231n: Convolutional Neural Networks for Visual Recognition (2017), http://cs231n.stanford.edu/2017/index.html

SqueezeNet (Iandola et al., 2017)

- AlexNet-level accuracy with 50x fewer parameters and <0.5Mb model size

- architecture:
  - squeeze layer(1x1 filters) feeds an expand layer(1x1 and 3x3 filters)

- benefits: memory footprints
  - model size: 510x smaller than AlexNet

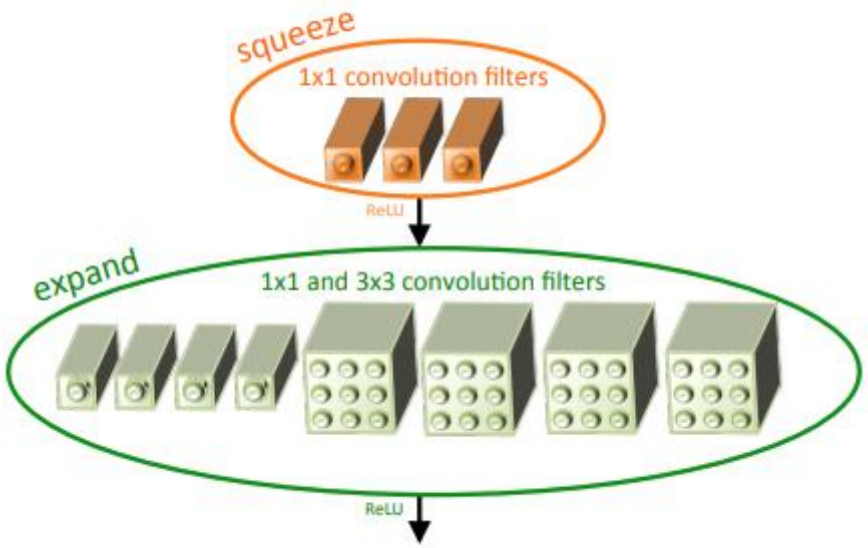


image source: Iandola, Forrest N., et al. "SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and <0.5MB model size" arXiv prepring arXiv:1602.07360 (2016)

MobileNet (Howard et al., 2017)

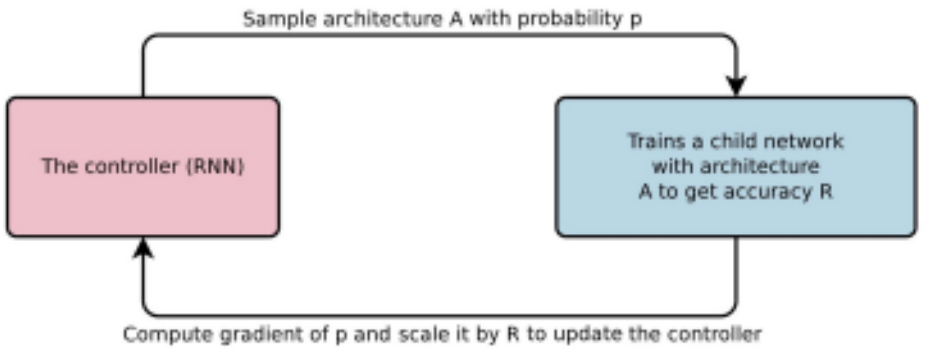
- efficient CNNs for mobile vision applications
  - uses depthwise separable conv -> more efficient (little accuracy loss)

- previous work (e.g. SqueezeNet): focused on reducing the number of parameters
  - more recent methods: reduce the number of operations and actual measured latency
- related approaches:
  - ShuffleNet (Zhang et al., 2017): group conv + channel shuffle operations
  - MobileNetV2 (Sandler et al., 2018): bottleneck + skip connection
  - MobileNetV3 (Howard et al., 2019): NAS based

image source: Google AI Blog

Neural architecture search (NAS) with RL (Zoph et al., 2016)

- a controller net (RNN)
  - learns to design a good architecture using REINFORCE (Williams, 1992)
  - a design ⟺ a string
  - reward ⟺ accuracy
- the controller net iterates:
  - 1. sample an arch from search space
  - 2. train the arch to get reward R
  - 3. compute the gradient of sample probability and scale it by R
  - 4. update controller parameters
  - increases likelihood of good arch
  - decreases likelihood of bad arch
- efficient NAS (ENAS; Pham et al., 2018): 1000x speed up



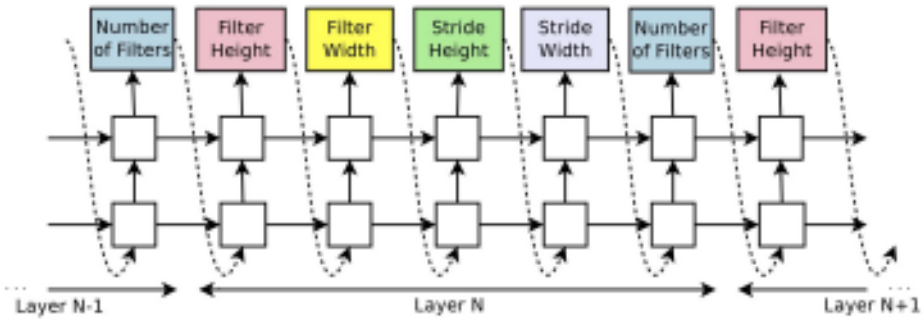


image source: Zoph, Barret, and Quoc V. Le. "Neural architecture search with reinforcement learning." arXiv preprint arXiv:1611.01578 (2016)

Google Cloud AutoML

image source: Google Cloud AutoML https://cloud.google.com/automl

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- famous four
  - AlexNet
  - VGG
  - GoogLeNet
  - ResNet
- improving ResNet
  - wide ResNet
  - ResNeXt
  - stochastic depth
  - SENet
- beyond ResNet
  - FractalNet
  - DenseNet
- other ideas
  - MobileNet
  - AutoML

- Rmk.
  - reasonable defaults: ResNet and SENet
  - many popular architectures available in model zoos
  - recent trends: meta-learning and autoML